Question. A market research firm reports that 35% of small businesses use paid social media advertising. A marketing researcher randomly selects 300 small businesses. Find the probability that the sample proportion of small businesses that use paid social media advertising is between 32% and 40%. Show all steps using the SPDC structure.

Detailed prompts

S – State:

1. Define the population (true) parameter in context.
2. Define the sample parameter in context.
3. What is 300? What does it represent?
4. What is 35%? What does it represent?
5. Write the problem (the probability you are asked to find using $\hat{p}$).

P – Plan:

1. Check randomness: Was the sample random? Why does randomness matter?
2. Check independence (10% condition): How many small businesses must there be in the population? Is $n < 10\%$ of N?
3. Check normality (large counts condition): Is the expected number of successes at least 10? Is the expected number of failures at least 10?
4. Compute the mean of the sampling distribution of $\hat{p}$.
5. Compute the standard deviation of the sampling distribution of $\hat{p}$, and write the Normal model.

D – Do:

1. Write the sampling distribution of the sample statistic.
2. Write the probability statement in terms of $\hat{p}$.
3. Write the probability statement in terms of Z formula.
4. Convert both boundaries to z-scores.

5. Use the standard Normal table and write the value of the probability.

C – Conclude:

1. Start your conclusion with "There is about a ___ probability that…"
2. What is the sample size, and who or what is in the sample?
3. What statistic does the sample produce?
4. What interval is in the question?
5. What does this proportion describe in context?